Chapter 2 Review

Campbell Biology, AP Edition, 12e

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Summary of Key Concepts

What are the characteristics of the chemical elements common to the molecular building blocks that are needed to capture and store energy in biological systems? **(BIG IDEAS 2)**

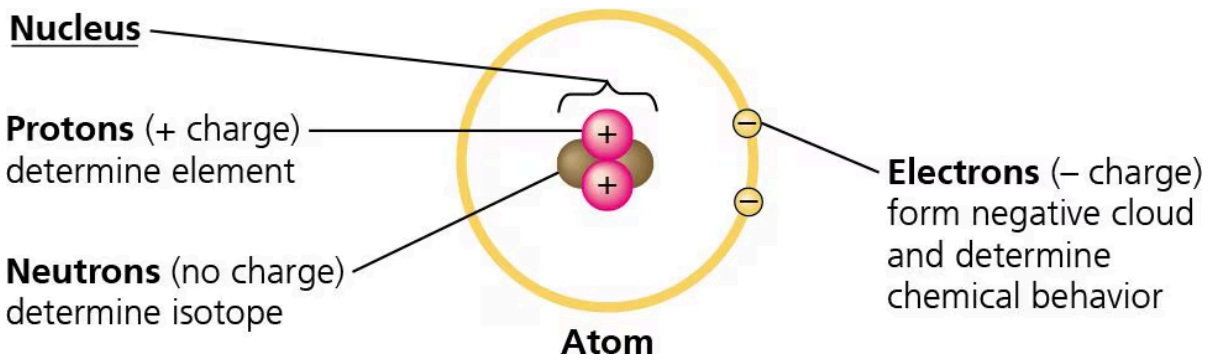
Concept 2.1: Matter consists of chemical elements in pure form and in combinations called compounds

- **Elements** cannot be broken down chemically to other substances. A **compound** contains two or more different elements in a fixed ratio. Oxygen, carbon, hydrogen, and nitrogen make up approximately 96% of living matter.

Compare an element and a compound.

Concept 2.2: An element's properties depend on the structure of its atoms

- An **atom**, the smallest unit of an element, has the following components:



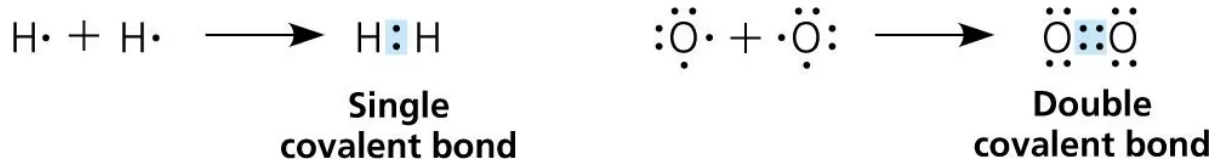
An atom is shown with a nucleus including protons (positive charge) which determine element and neutrons (negative charge) which determine isotope. Electrons (negative charge) form negative cloud and determine chemical behavior.

- An electrically neutral atom has equal numbers of electrons and protons; the number of protons determines the **atomic number**. The **atomic mass** is measured in **daltons and is roughly equal to the mass number**, the sum of protons plus neutrons. **Isotopes** of an element differ from each other in neutron number and therefore mass. Unstable isotopes give off particles and energy as radioactivity.
- In an atom, electrons occupy specific **electron shells**; the electrons in a shell have a characteristic energy level. Electron distribution in shells determines the chemical behavior of an atom. An atom that has an incomplete outer shell, the **valence shell**,****is reactive.
- Electrons exist in ****orbitals**, three-dimensional spaces with specific shapes that are components of electron shells.

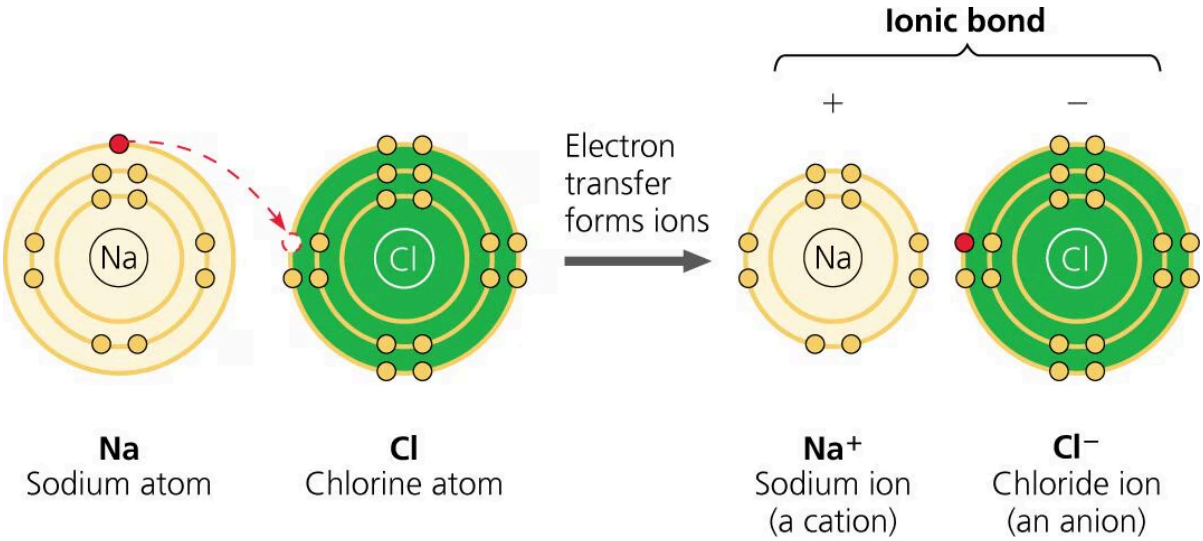
DRAW IT*Draw the electron distribution diagrams for neon*(N10e)*and argon*(A18r). *Use these diagrams to explain why these elements are chemically unreactive.*

Concept 2.3: The formation and function of molecules and ionic compounds depend on chemical bonding between atoms

- ****Chemical bonds** form when atoms interact and complete their valence shells. ****Covalent bonds** form when pairs of electrons are shared:



- Part a. Single covalent bond: Two hydrogen atoms, each with a single electron in the outer shell, form a single covalent bond, sharing their one pair of electrons.
- Part b. Double covalent bond: Two oxygen atoms, each with six electrons in the outer shell, share two pairs of electrons in forming a double covalent bond.
- ********Molecules** consist of two or more covalently bonded atoms. The attraction of an atom for the electrons of a covalent bond is its **electronegativity**. If both atoms are the same, they have the same electronegativity and share a ****nonpolar covalent bond**. Electrons of a ****polar covalent bond** are pulled closer to the more electronegative atom, such as the oxygen in H2O.
- An ****ion** forms when an atom or molecule gains or loses an electron and becomes charged. An **ionic bond** is the attraction between two oppositely charged ions:



The lone valence electron of a sodium atom with one electron in the third shell, eight electrons in the second shell, and two electrons in the first shell, is transferred to join the 7 valence electrons in the third shell of a chlorine atom to form Sodium chloride (N a C l). Each resulting ion has a completed valence shell. An ionic

bond can form between the oppositely charged ions. N a plus has eight electrons in the second shell and two electrons in the first shell. C l minus has 8 electrons in the third shell, eight electrons in the second shell, and two electrons in the first shell.

- Weak interactions reinforce the shapes of large molecules and help molecules adhere to each other. A ******hydrogen bond** is an attraction between a hydrogen atom carrying a partial positive charge (δ+) and an electronegative atom carrying a partial negative charge (δ−). ******Van der Waals interactions** occur between transiently positive and negative regions of molecules.
- A molecule's shape is determined by the positions of its atoms' valence orbitals. Covalent bonds result in hybrid orbitals, which are responsible for the shapes of H_2O, CH_4, and many more complex biological molecules. Molecular shape is usually the basis for the recognition of one biological molecule by another.

In terms of electron sharing between atoms, compare nonpolar covalent bonds, polar covalent bonds, and the formation of ions.

Concept 2.4: Chemical reactions make and break chemical bonds

- ******Chemical reactions** change ******reactants** into ******products** while conserving matter. All chemical reactions are theoretically reversible. ******Chemical equilibrium** is reached when the forward and reverse reaction rates are equal.

What would happen to the concentration of products if more reactants were added to a reaction that was in chemical equilibrium? How would this addition affect the equilibrium?